Controllable Universal Fair Representation Learning

Yue Cui
The Hong Kong University of Science and Technology, Hong Kong SAR, China
ycuias@cse.ust.hk

Chen Ma
City University of Hong Kong, Hong Kong SAR, China
chenma@cityu.edu.hk

Kai Zheng
University of Electronic Science and Technology of China, Chengdu, China
zhengkai@uestc.edu.cn

Lei Chen
The Hong Kong University of Science and Technology, Hong Kong SAR, China
leichen@cse.ust.hk

Xiaofang Zhou
The Hong Kong University of Science and Technology, Hong Kong SAR, China
zxf@cse.ust.hk

ABSTRACT

Learning fair and transferable representations of users that can be used for a wide spectrum of downstream tasks (specifically, machine learning models) has great potential in fairness-aware Web services. Existing studies focus on debiasing *w.r.t.* a small scale of (one or a handful of) fixed pre-defined sensitive attributes. However, in real practice, downstream data users can be interested in various protected groups and these are usually not known as prior. This requires the learned representations to be fair *w.r.t.* all possible sensitive attributes. We name this task universal fair representation learning, in which an exponential number of sensitive attributes need to be dealt with, bringing the challenges of unreasonable computational cost and un-guaranteed fairness constraints. To address these problems, we propose a controllable universal fair representation learning (CUFRL) method. An effective bound is first derived via the lens of mutual information to guarantee parity of the universal set of sensitive attributes while maintaining the accuracy of downstream tasks. We also theoretically establish that the number of sensitive attributes that need to be processed can be reduced from exponential to linear. Experiments on two public real-world datasets demonstrate CUFRL can achieve significantly better accuracy-fairness trade-off compared with baseline approaches.

CCS CONCEPTS

• **Information systems** → **Personalization**; • **Mathematics of computing** → *Information theory*.

KEYWORDS

Representation learning, Fairness, Demographic parity

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1 INTRODUCTION

Machine learning models developed for intelligent Web applications such as personalized search, online advertising and hiring platforms usually utilize user data with rich demographic features. Unfairness and biases that already exist in real-world data may be involuntarily introduced to these models. Learned from data with ethical implications, there is a piece of growing evidence that these algorithms may unintentionally provide discriminate predictions [25, 30, 44, 47]. Therefore, machine learning debiasing is an important task in Web.

Fairness-aware user representation learning allows data holders in Web to release user data to third parties without prior knowledge of the tasks performed (specifically, machine learning models that use these representations as inputs) [48], while satisfying the fairness constraints and preserving data utility. In a typical fair representation learning setting, only fairness on a single (or a handful of) specified sensitive attribute is considered [12, 13, 48, 55]. However, problems arise when not only the downstream tasks, but the debiasing attributes of data users' interests are unknown. For example, in a hiring web platform, institutions looking for STEM-related job seekers might be interested in debiasing on sensitive attribute *gender* while institutions looking for politics-related employers might be interested in the sensitive attribute *age*. To meet the various debiasing targets of data users, it yields a demand for learning representations that are fair over a variety of sensitive attributes. More specifically, the universal set of all singular sensitive attributes and their possible combinations (see Definition 1).

We refer to this problem as universal fair representation learning. More specifically, given a user dataset where each user is characterized by a set of attributes, which may include both sensitive and non-sensitive ones, we aim to learn user representations that satisfy certain fairness constraints across all sensitive attributes and their possible combinations (see Definition 1). Note that the above problem formulation is based on group fairness. In group fairness, each sensitive attribute defines a demographic partition of the population, which is in the form of several groups (*e.g.,* the sensitive attribute "gender" divides the population into groups "male" and "female"), and a group fairness constraint (*e.g.,* demographic parity, see [10] or Definition 2) should be satisfied across all groups in that partition. The problem of universal fair representation learning brings non-trivial challenges.

First, **fairness** needs to be maintained over an exponential number of sensitive attributes. Even in the simplest case, assuming there are m binary sensitive attributes, there are $C_m^1 + ... + C_m^m = 2^m - 1$ attributes and the i-th one $i \in [1, ..., m]$ has 2^i different values. Since in fair representation learning, debiasing is usually achieved by adding a constraint per attribute [13, 23], the computational cost to process such an enormous number of constraints is unreasonable. It is worth mentioning that works in *subgroup fairness* [2, 24] deal with a similar problem of exponential/infinite collection of protected groups. The mainstream solutions follow auditing and learning fashion, in which an auditor monitors the worst subgroup and the learner makes adaptive model updates accordingly. However, tailored on the pre-defined auditing and learning algorithm, the detecting-debiasing paradigm cannot guarantee utility and fairness are transferable to downstream tasks (specifically, machine learning models). Moreover, their definition of fairness concerning individual subgroup *v.s.* the whole population/the governed majority [25] also distinguishes us.

Second, the learned representations need to be as **expressive** as possible. It has been widely acknowledged that there are compromises between accuracy and fairness [13, 25]. It is desirable to not have a severe negative effect on accuracy when fairness constraints *w.r.t.* massive sensitive attributes are to be met.

To this end, in this paper, we propose an information estimation based bi-level optimization framework, named CUFRL, aiming at learning data representations that satisfy universal group fairness under demographic parity while maximizing representation expressiveness. We first extend attribute-wise demographic parity definition to the universal-set setting and prove the disparity can be constrained in a provable fashion via the lens of mutual information between the learned representation and sensitive attributes. To tackle the challenge of the massive number of sensitive attributes, we show that the number of sensitive attributes of concern can be reduced to a linear number via the property of independence and mutual information, which indicates the problem can be achieved by controlling the bias of only a linear number of sensitive attributes. Further, provable upper/lower bounds are obtained to estimate the mutual information terms. We instantiate the estimations by leveraging matured tools and objectives such as maximum likelihood, Kullback–Leibler loss, and generative adversarial learning.

Our contributions can be summarized as follows:

- We introduce a practical setting, named universal fair representation learning. It allows data providers to release debiased user data that is expressive and fair over all sensitive attributes in a given dataset. This is important considering the interested protected groups of downstream data users are various and unknown in advance.
- An end-to-end and guaranteed solution CUFRL is proposed via the lens of mutual information. CUFRL is model-agnostic with the characterization independent of downstream tasks (specifically, machine learning models), which enables a wide application scenario. In addition, to make the computational cost reasonable, instead of directly dealing with an exponential number of combined sensitive attributes, we draw connections between levels of attributes and reduced the to-be-processed attributes to a linear number.
- Extensive experiments on two public real-world datasets verify that compared with baselines, CUFRL achieves significantly better accuracy-fairness trade-off.

Table 1: Summary of Notations.

Notations	Descriptions
$\mathbf{c}_i$, n_i	A sensitive attribute and its total number of categories
$\mathbf{C}^l$	Collection of level l combined sensitive attribuets
$\mathbf{C}^U$	Collection of the universal set of sensitive attributes
$\mathcal{A}$	A learning algorithm
$\Delta_{DP}(\mathcal{A}, \mathbf{c}_k)$	Demographic disparity on $\mathcal{A}$, sensitive attribute $\mathbf{c}_k$
$\Delta_{DP}(\mathcal{A}, \mathbf{C}^U)$	Universal demographic disparity on $\mathcal{A}$, universal sensitive attribute set $\mathbf{C}$
$\mathbf{x}, \hat{\mathbf{y}}, \mathbf{y}$	Input features, predicted outcome, and groundtruth label
$\mathbf{z}$	Learned representation vector
$\mathbf{I}(p; q)$	Mutual information between distributions p and q
β	Tune-able parameter to control the trade-off between fairness and accuracy

2 PROBLEM FORMULATION

In the section, we first introduce the definition of combined sensitive attributes, and then, the measurement of demographic parity under universal set setting is proposed. At last, we formulate the problem.

2.1 Notations and Definitions

Key notations used in this paper are summarized in Table 1.

Denote the sensitive attributes in a given dataset as singular sensitive attributes (*e.g.,* gender, race), selecting any number of attributes from the singular ones, one can form a combined sensitive attribute (*e.g.,* gender-race), denoted as level-l sensitive attribute, ($l \leq 2$). Adopting the widely used assumption that each attribute can be simplified as categorical [34], given m singular sensitive attributes $\mathbf{C}^1 = \{\mathbf{c}_1, ..., \mathbf{c}_m\}$, where $\mathbf{c}_i \in \{1, ..., n_i\}$, n_i the total number of categories (values) of $\mathbf{c}_i$, we define a level-l sensitive attribute as follows.

Definition 1. ***Level-l Sensitive Attribute***: *Selecting l attributes from $\mathbf{C}^1$, where $l \leq m$, denoted as $\{\mathbf{c}_{i,1}, ..., \mathbf{c}_{i,l}\}$, we can form a level-$l$ sensitive attribute that has $\prod_{j=1}^{l} n_{i,j}$ categories in total, where $n_{i,j}$ denotes the number of categories of singular sensitive attribute $\mathbf{c}_{i,j}$. Each of the new categories denotes a mixture formed by picking up one category value from each of the sensitive attribute in $\{\mathbf{c}_{i,1}, ..., \mathbf{c}_{i,l}\}$.*

For example, if there are two singular sensitive attributes $\mathbf{c}_1 \in \{a, b\}$ and $\mathbf{c}_2 \in \{c, d\}$, where a, b are categories of $\mathbf{c}_1$, then there will be one and only one combined sensitive attribute, *i.e.,* the level-2 sensitive attribute $\mathbf{c}_{12} \in \{(a, c), (a, d), (b, c), (b, d)\}$.

The level-1 combined sensitive attributes are denoted as the singular sensitive attributes themselves. For m singular attributes,

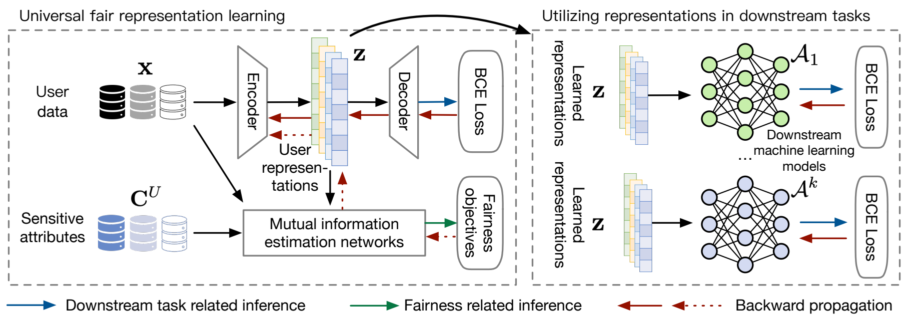


Figure 1: Overview of CUFRL.

there are in total, m levels of combined sensitive attributes. We denote the collection of all m levels of sensitive attributes as $\mathbf{C}^U$, which can be described as the universal set of sensitive attributes and U stands for universal. If not specified, for simplicity, the notion of "sensitive attribute" in following sections can refer to any level of attributes in $\mathbf{C}^U$. In universal fair representation learning, the learned representation is required to be fair *w.r.t.* all sensitive attributes in $\mathbf{C}^U$.

The widely used fairness criterion demographic parity (DP) can be achieved by reducing the maximum disparity of groups defined by a sensitive attribute to zero. Formally,

Definition 2. ***(Attribute-wise) Demographic Parity (DP):*** *[10] Given any learning algorithm $\mathcal{A}$, a sensitive attribute $\mathbf{c}_k$, $\mathcal{A}$ has demographic parity* w.r.t. $\mathbf{c}_k$ *if:*

$$\Delta_{DP}(\mathcal{A}, \mathbf{c}_k) = \max_{i,j} |P(\hat{\mathbf{y}} = 1|\mathbf{c}_k = i) - P(\hat{\mathbf{y}} = 1|\mathbf{c}_k = j)| = 0 \tag{1}$$

*where $\mathcal{A}$ can be any downstream machine learning models (*aka. *downstream tasks), $\hat{\mathbf{y}}$ denotes outcome of $\mathcal{A}$, P is probability, $i, j \in [1, ..., n_k]$, and n_k is the total number of categories of $\mathbf{c}_k$.*

Based on the attribute-wise DP, we can define the following demographic parity for a universal set of sensitive attributes.

Definition 3. ***Universal Demographic Parity (UDP):*** *Given the collection of all sensitive attributes,* i.e., $\mathbf{C}^U$*, an algorithm follows universal demographic parity* w.r.t. $\mathbf{C}^U$ *if:*

$$\Delta_{DP}(\mathcal{A}, \mathbf{C}^U) = ||\mathbf{v}_{\Delta_{DP}}||_2 = \sqrt{\sum_{i=1}^{|\mathbf{C}^U|} \Delta_{DP}(\mathcal{A}, \mathbf{c}_i)^2} = 0$$

where $|\mathbf{C}^U|$ denotes the number of variables in $\mathbf{C}^U$, $\mathbf{v}_{\Delta_{DP}} = [\Delta_{DP}(\mathcal{A}, \mathbf{c}_1)$ $, ..., \Delta_{DP}(\mathcal{A}, \mathbf{c}_{|\mathbf{C}^U|})] \in \mathbb{R}^{|\mathbf{C}|}$ is a vector formed by attribute-wise Δ_{DP}, and $||\cdot||_2$ is the L2 norm.

Definition3 serves as an simple upper bound of attribute-wise Δ_{DP}. That is to say, when $\Delta_{DP}(\mathcal{A}, \mathbf{C}^U)$ is 0, each element in $\mathbf{v}$ must also be 0. Though L1 and others norms can also characterize the volume of disparity concerning all sensitive attributes and serve as an upper bound, we will show in later sections that L2 norm possesses good properties and facilitates proofs of some key theorems.

2.2 Problem Statement

Equipped with the above preliminaries, the studied problem can be formulated as follows:

Controllable Universal Fair Representation Learning: Denote the dataset as $\mathcal{D} = \{x_i, y_i, c_i\}_{i=1}^n$, where values of x_i, y_i, c_i are *i.i.d.* realizations of random variables $\mathbf{x}, \mathbf{y}, \mathbf{c}$ distributed by $p(\mathbf{x}, \mathbf{y}, \mathbf{c})$, $\mathbf{x}$ are features, $\mathbf{y}$ is the label, and $\mathbf{c}$ are singular demographic sensitive attributes, where $x_i \in \mathbb{R}^n, y_i \in \mathbb{R}^1, c_i \in \mathbb{R}^m$, $\mathbf{c}$ may all or partially be included in $\mathbf{x}$. Suppose we have an encoding network NN_{en}, which embeds the input features $\mathbf{x}$ into a representation vector $\mathbf{z}$. Denote that $\mathbf{z}, \mathbf{x} \sim p(\mathbf{z}|\mathbf{x})$. The goal is to learn fair representations $\mathbf{z}$ from $\mathcal{D}$ such that for any learning algorithm $\mathcal{A}$ (*aka.* a downstream task) that acts on $\mathbf{z}$, UDP can be achieved while the expressiveness of the representation is maximized.

3 THE CONTROLLABLE UNIVERSAL FAIR REPRESENTATION LEARNING METHOD

In this section, we first propose the objectives used in CUFRL and then introduce their practical instantiations. An overview of the proposed CUFRL is illustrated in Figure 1.

3.1 Objective Deriving

Similar to other fair representation learning tasks, universal fair representation learning has two goals: 1) maximizing data utility, and 2) minimizing disparity. Interpreted in the principle of information [50], since $\mathbf{z}$ is the trainable part of the framework, the first goal can be straightforwardly placed as: it requires the representation $\mathbf{z}$ to maximize the mutual information between itself and $\mathbf{y}$ (the label), *i.e.,* maximizing $I(\mathbf{y}; \mathbf{z})$, which is equivalent to minimizing $-I(\mathbf{y}; \mathbf{z})$. However, **it is unclear how disparity, *i.e.,* $\Delta_{DP}(\mathcal{A}, \mathbf{C}^U)$ can be related**.

The majority of early studies model the fairness criterion conceptually in an in-process fashion and evaluate related metrics after training, in which there could be a gap between the debiasing target and fairness measurement. Recently, provable guarantees are proposed to bridge such gaps [33], where mutual information has been proven to be promising to serve as a link between input features, intermediate representations and demographic parity [12, 13]. Inspired by these works, we here show that the sum of mutual information between $\mathbf{z}$ and $\mathbf{c}_i$ can be used to bound the disparity $\Delta_{DP}(\mathcal{A}, \mathbf{C}^U)$.

Theorem 4. *For $z, c_i \sim p(\mathbf{z}, \mathbf{c}_i)$, $z \in \mathbb{R}^d$, $\mathbf{c}_i \in \mathbf{C}^U$, $c_i \in \{1, ..., n_i\}$, and any decision algorithm $\mathcal{A}$ that acts on $\mathbf{z}$, we have*

$$\sum_{i=1}^{|\mathbf{C}^U|} I(\mathbf{z}; \mathbf{c}_i) \geq g(\gamma, \Delta_{DP}(\mathcal{A}, \mathbf{C}^U))$$

where $I(\mathbf{p}; \mathbf{q})$ denotes the mutual information between two probability distributions $\mathbf{p}$ and $\mathbf{q}$, $\gamma = \min_i \alpha_i$, $\alpha_i = \min_j \pi_{i,j}$, $\pi_{i,j} = P(\mathbf{c}_i = j)$, and $g(x, y) = xy^2$.

The proof of Theorem 4 is delayed to Appendix A.

Proving the disparity can be limited by mutual information, we can result in the following objective:

$$O' \triangleq \min_{p(\mathbf{z}|\mathbf{x})\in\Omega}[-I(\mathbf{y};\mathbf{z}) + \beta' \sum_{i=1}^{|\mathbf{C}^U|} I(\mathbf{z};\mathbf{c}_i)] \tag{2}$$

that is, maximizing the mutual information between $\mathbf{y}$ and $\mathbf{z}$ to achieve good performance on the prediction task while minimizing the sum of mutual information between $\mathbf{z}$ and $\mathbf{c}_i$ to achieve good parity over $\mathbf{C}^U$. Ω denotes the search space of the conditional distribution of $\mathbf{z}$ given $\mathbf{x}$ and β' is a hyper-parameter to control the trade-off between the two sub-objectives.

Given that all sensitive attributes are considered, it could be extremely expensive when parameterizing the distributions, due to that such a process is usually achieved of complexity in proportion to the number of values of attributes [13]. By Theorem 5, we show how to overcome this difficulty by only considering a linear number of sensitive attributes.

To simplify the description, we denote the combination of sensitive attributes $\mathbf{c}_i$ and $\mathbf{c}_j$ as $\mathbf{c}_{ij}$. Given sensitive attributes are demographic statistics, we adopt the assumption that level-1 sensitive attributes are independent [1, 34].

Theorem 5. *For a combined sensitive attribute $\mathbf{c}_{ij}$, which is formed by two independent sensitive attribute $\mathbf{c}_i$ and $\mathbf{c}_j$, where $c_i \in \{1, .., n_i\}$, $c_j \in \{1, ..., n_j\}$, we have: If $I(\mathbf{z};\mathbf{c}_i) = 0$ and $I(\mathbf{z};\mathbf{c}_j) = 0$, then $I(\mathbf{z};\mathbf{c}_{ij}) = 0$ is an essential equivalence.*

Proof. Mutual information being zero is equivalent to the independence of two random variables. Then if $\mathbf{c}_i$ is independent of $\mathbf{z}$, $\mathbf{c}_j$ is independent of $\mathbf{z}$, and $\mathbf{c}_i$, $\mathbf{c}_j$ are independent, then $\mathbf{c}_{ij}$ is essentially independent of $\mathbf{z}$ [14, 49]. This completes the proof. □

That is, to minimize the mutual information between $\mathbf{z}$ and a sensitive attribute, we can minimize that between $\mathbf{z}$ and a lower level sensitive attribute. Since we can always construct higher-level sensitive attributes by adding one level-1 attribute recursively, we only need to focus on level-1 sensitive attributes. Based on Theorem 5, we can rewrite the objective as follows

$$O \triangleq \min_{p(\mathbf{z}|\mathbf{x})\in\Omega}[-I(\mathbf{y};\mathbf{z}) + \beta \sum_{i=1}^{|\mathbf{C}^1|} I(\mathbf{z};\mathbf{c}_i)] \tag{3}$$

where $\mathbf{C}^1$ is the collection of all level-1 combined sensitive attributes, *i.e.,* the singular sensitive attributes.

3.2 Mutual Information Estimation

Despite being a pivotal tool linking parity and intermediate representation, mutual information has historically been difficult to compute [39], especially when the probability distributions of data are unknown [4]. As a common practice, we here adopt provable lower/upper bounds as estimations and use existing objectives to instantiate them.

3.2.1 Estimate $I(\mathbf{y};\mathbf{z})$. According to [38, 43], for any random variables $\mathbf{a}$ and $\mathbf{b}$ and any function $f(\mathbf{a},\mathbf{b}) \in \mathbb{R}$, we have

$$I(\mathbf{a};\mathbf{b}) \geq \mathbb{E}_{p(\mathbf{a},\mathbf{b})}[f(a,b)] - \mathbb{E}_{p(\mathbf{a})p(\mathbf{b})}[exp(f(a,b)-1)] \tag{4}$$

where $\mathbb{E}$ denotes expectation.

Following a similar method in [52] we estimate upper $I(\mathbf{y};\mathbf{z})$ as

Proposition 6. ***The lower bound for** $I(\mathbf{y};\mathbf{z})$**:** Apply the above formulation on $(\mathbf{y},\mathbf{z})$ and set $f(\mathbf{y},\mathbf{z}) = 1 + log\frac{p(y|z)}{p(y)p(y)}$, we have*

$$I(\mathbf{y};\mathbf{z}) \geq 1 + \mathbb{E}_{p(\mathbf{y},\mathbf{z})}[log\frac{p(y|z)}{p(y)}] + \mathbb{E}_{p(\mathbf{y})p(\mathbf{z})}[\frac{p(y|z)}{p(y)}]. \tag{5}$$

To characterize the above equation, we approximate $p(y|z)$ with $q(y|x) = NN_{de}(z)$, where NN_{de} denotes a parametrized neural network to decode z and get the prediction of y, and estimating $p(y)$ from the distribution of the data. To this end, ignoring constants, the RHS of Equation 5 can be expressed as cross entropy of the classification task. We implement NN_{de} as a 1-layer MLP.

3.2.2 Estimate $\sum_{i=1}^{|\mathbf{C}^1|} I(\mathbf{z};\mathbf{c}_i)$. Since it can be derived that for any three random variables $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$, $I(\mathbf{a};\mathbf{b}) + I(\mathbf{a};\mathbf{c}|\mathbf{b}) = I(\mathbf{a};\mathbf{c}) + I(\mathbf{a};\mathbf{b}|\mathbf{c})$, as proposed in [13], the second term of Equation 3 can be re-written as:

$$\begin{aligned}\sum_{i=1}^{|\mathbf{C}^1|} I(\mathbf{z};\mathbf{c}_i) &= \sum_{i=1}^{|\mathbf{C}^1|} I(\mathbf{z};\mathbf{c}_i|\mathbf{x}) + I(\mathbf{z};\mathbf{x}) - I(\mathbf{z};\mathbf{x}|\mathbf{c}_i) \\ &= |\mathbf{C}^1| I(\mathbf{z};\mathbf{x}) - \sum_{i=1}^{|\mathbf{C}^1|} I(\mathbf{z};\mathbf{x}|\mathbf{c}_i)\end{aligned}$$

where $I(\mathbf{z};\mathbf{c}_i|\mathbf{x}) = 0$ because $\mathbf{z}$ is independent of $\mathbf{c}_i$ given $\mathbf{x}$, $\forall \mathbf{c}_i \in \mathbf{C}^1$. We can thus bound $I(\mathbf{z};\mathbf{c}_i)$ using the upper bound of $I(\mathbf{z};\mathbf{x})$, which is the information bottleneck [50] term, and the lower bound of $I(\mathbf{z};\mathbf{x}|\mathbf{c}_i)$, aiming at maximizing the information between $\mathbf{z}$ and $\mathbf{x}$ without $\mathbf{c}_i$.

Estimate $I(\mathbf{z};\mathbf{x})$. Follow a similar fashion as [13, 43], we derive the upper bound of $I(\mathbf{z};\mathbf{x})$ as

$$\begin{aligned}I(\mathbf{z};\mathbf{x}) &= \mathbb{E}_{p(\mathbf{z},\mathbf{x})} log\frac{p(z|x)}{q(z)} - KL(p(\mathbf{z})\|q(\mathbf{z})) \\ &\leq \mathbb{E}_{p(\mathbf{z},\mathbf{x})} log\frac{p(z|x)}{q(z)} = \mathbb{E}_{p(\mathbf{x})} KL\big(p(\mathbf{z}|\mathbf{x})\|q(\mathbf{z})\big)\end{aligned} \tag{6}$$

where $p(\mathbf{z})$ is any distribution, $KL(\cdot||\cdot)$ is the Kullback–Leibler (KL) divergence loss [28]. To characterize Equation 6, we may simply set $q(\mathbf{z})$ as $\mathcal{N}(0, I)$, and estimate $p(z|x)$ as $q(z|x) = NN'_{en}(x)$, where NN'_{en} is implemented as a 1-layer MLP. KL loss can thus be obtained to estimate Equation 6.

Estimate $\sum_{i=1}^{|\mathbf{C}^1|} I(\mathbf{z};\mathbf{x}|\mathbf{c}_i)$. Applying the Nguyen, Wainwright & Jordan's bound [38] to random variables $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$, we have

$$I(\mathbf{a};\mathbf{b}|\mathbf{c}) \geq \mathbb{E}_{p(\mathbf{a},\mathbf{b},\mathbf{c})}[g(a,b,c)] - e^{-1}\mathbb{E}_{p(\mathbf{a}|\mathbf{c})p(\mathbf{b},\mathbf{c})}[\exp g(a,b,c)] \tag{7}$$

Proposition 7. ***The lower bound for** $I(\mathbf{z};\mathbf{x}|\mathbf{c}_i)$**:** Apply Equation 7 on $(\mathbf{z},\mathbf{x},\mathbf{c}_i)$, plugging in $g(z,x,c_i) = 1 + log\frac{p(z,x,c_i)}{p(z|c_i)p(x,c_i)}$, we can obtain that*

$$\begin{aligned}I(\mathbf{z};\mathbf{x}|\mathbf{c}_i) \geq 1 + \mathbb{E}_{p(\mathbf{z},\mathbf{x},\mathbf{c}_i)}[log\frac{p(z,x,c_i)}{p(z|c_i)p(x,c_i)}] - \\ \mathbb{E}_{p(\mathbf{z}|\mathbf{c}_i)p(\mathbf{x},\mathbf{c}_i)}[log\frac{p(z,x,c_i)}{p(z|c_i)p(x,c_i)}].\end{aligned} \tag{8}$$

Though $g(z,x,c_i) = 1 + log\frac{p(z,x,c_i)}{p(z|c_i)p(x,c_i)}$ is an optimal choice to make the bound tight, it can be challenging to compute the joint density [37]. We here adopt generator-discriminator based method [35, 37] to empirically estimate the mutual information.

The key idea is to use trainable neural networks to discriminate samples from the joint distribution $p(\mathbf{z}, \mathbf{x}, \mathbf{c}_i)$ and product distribution $p(\mathbf{z}|\mathbf{c}_i)p(\mathbf{x}, \mathbf{c}_i)$. In our implementation, we use a 1-layer MLP to characterize the distribution of $p(\mathbf{z}|\mathbf{c}_i)$, *i.e.,* $p(\mathbf{z}|\mathbf{c}_i) = NN_G(\mathbf{c}_i)$, while $p(\mathbf{z}, \mathbf{x}, \mathbf{c}_i)$ and $p(\mathbf{x}, \mathbf{c}_i)$ are estimated from the data. The sampled instances from the two distributions can be denoted as $\mathcal{B}_{joint}$ and $\mathcal{B}_{prod}$, with labels **1** and **0**, respectively, where the bold numbers denote vectors. Then a discriminator, denoted as NN_D, is trained on $\{\mathcal{B}_{joint}, \mathbf{1}\}$ and $\{\mathcal{B}_{prod}, \mathbf{0}\}$ with cross entropy loss. NN_D is also implemented as a 1-layer MLP. Mathematically, suppose we sample K instances from $\{\mathcal{B}_{joint}, \mathbf{1}\}$ and $\{\mathcal{B}_{prod}, \mathbf{0}\}$ separately, the loss function can be formulated as:

$$L_{c_i} = -\frac{1}{2K}\Big[\sum_{(z,x,c_i)\in\mathcal{B}_{joint}} log\omega(z, x, c_i) + \sum_{(z,x,c_i)\in\mathcal{B}_{prod}} log\big(1 - \omega(z, x, c_i)\big) \tag{9}$$

where $\omega(z, x, c_i)$ is the prediction of NN_D *w.r.t.* the probability of whether the sample is from the joint distribution. In other words, the optimal $\omega(z, x, c_i)$ approximates $p(z, x, c_i)$ and minimizing Equation 9 will help the parameterized distributions close to true ones. We then sum over L_{c_i} to obtain L_c, which serves as an estimation of conditional mutual information $\sum_{i=1}^{|\mathbf{C}^1|} I(\mathbf{z}; \mathbf{x}|\mathbf{c}_i)$. Due to limited space, we defer the pseudo code of computing L_c to Appendix B.

4 EXPERIMENT AND RESULTS

4.1 Experimental Setup

4.1.1 Datasets and Metrics. We conduct experiments on two benchmark datasets **Heritage Health** [1] and **AdultCensus** [27]. The Heritage Health dataset is collected by Heritage Provider Network, containing medical records of its member patients. The goal is to predict the Charleson Index for each patient, which indicates the ten-year mortality. The AdultCensus dataset is part of the 1994 census data of the US. The goal is to predict if a person makes more than 50K per year. Following the same procedure as [13, 36, 48], statistics of the pre-processed dataset are described in Table 2. We compute the F1 score to measure the performance of the classification task and $\Delta_{DP}(\mathcal{A}, \mathbf{C}^U)$ to measure UDP.

Table 2: Statistics of datasets.

Dataset	# samples	# feat.	# singular sensitive attr.
Heritage Health	55924	123	3
AdultCensus	45222	99	14

4.1.2 Implementation Details. All experiments are implemented on one Intel (R) Xeon(R) Gold 6278C CPU @ 2.60GHz and one NVIDIA GeForce RTX 2080 GPU. Each dataset is split into 6/2/2 for training, validation, and test. We fix the batch size to 128, following [13]. For representation learning, the weight decay rate is set as 1e-4. The embedding dimension of $\mathbf{z}$ is set as 8 for the Heritage Health dataset and 16 for the AdultCensus dataset. We initialize the learning rate as 1e-3 and decay 0.01 every 10 epochs. In order to verify if the learned representation is model-agnostic, four decision algorithms (downstream tasks) are evaluated: Logistic Regression, 2-layer MLP, Random Forest, and AdaBoost. Specifically, the learned representations of training, validation, test datasets are then used as inputs for the training, validation and testing of these machine learning algorithms, where only the classification task loss is deployed. The maximum number of iterations for Logistic Regression and MLP is set as 1000, and apart from that other hyper-parameter settings are the same as they are default in scikit-learn [2].

We mainly consider two training paradigms to obtain fair representations: 1) pre-training based and 2) joint-training based. For the pre-training based approach, the network is optimized with the objective of minimizing classification loss only for the first few epochs before Equation 3 taking over. For the joint-training based approach, Equation 3 will be applied from the very beginning of training. Unless otherwise specified, the default paradigm is joint-training.

4.1.3 Comparison Baselines. To the best of our knowledge, there is no previous model intentionally designed for universal fair representation learning. *Subgroup fairness* works are the most related for that they also deal with a large number protected groups. Therefore, we adapt the auditing and learning (AL) based approach proposed in [24] on existing fair representation methods with modifications on the fairness constraints and set target debiasing attributes as $\mathbf{C}^U$. Note it is the fair representation learning network set as base model for auditing and learning but not the downstream machine learning models.

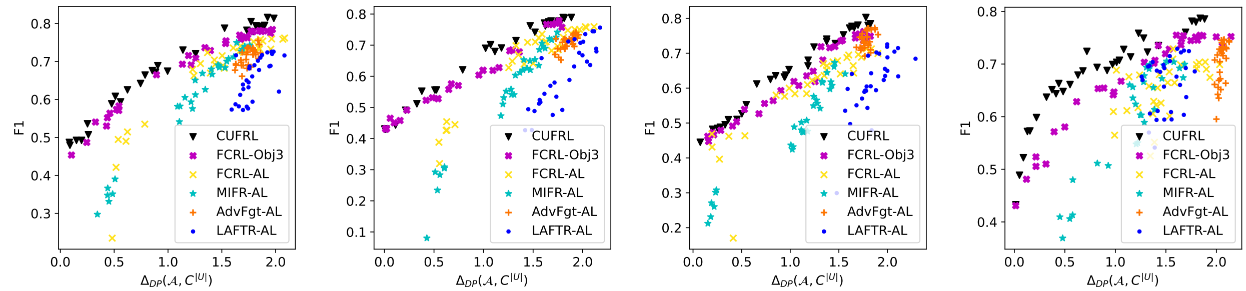
Denote models equipped with auditing and learning as [model-name]-AL, we have four related baselines, including information theory based models: **FCRL-AL** [13], which uses contrastive information estimators for fair representation learning; and **MIFR-AL** [48], an compatible mutual information based fair representation learning method; and state-of-the-art adversarial-based models **Adversarial Forgetting-AL (AdvFgt-AL)** [19], and **LAFTR-AL** [32], which minimizes a more directed adversarial approximation to parity. The structure and hyper-parameter settings of all the baselines are the same as [13]. Note that LAFTR is applicable to binary attributes, so we only deploy it on the AdultCensus dataset. Since our work is also closely related to state-of-the-art work FCRL [13], we also include a baseline **FCRL-Obj3** by directly changing objective function of FCRL to Equation 3 for comparison.

4.2 Accuracy and Fairness Trade-off

We measure the trade-off between F1-$\Delta_{DP}(\mathcal{A}, \mathbf{C}^U)$ by tuning trade-off related hyper-parameters of the models, *i.e.,* β in our notations. The results on AdultCensus dataset are presented in Figure 3 and results on Health Heritage are deferred to Appendix C due to limited space. It can be observed in Figure 2 that CUFRL surpasses all compared methods with a large margin. At similar parity, CUFRL achieves ~4% higher F1 on the AdultCensus dataset than the best baseline, and up to 0.2 lower disparity at a similar F1 score. AL-based methods are shown to be not able to achieve high F1 and low disparity. This because the characterization of auditor and learner are heavily algorithm-dependent, which makes the learned representation not applicable to unknown downstream prediction

[1] https://www.kaggle.com/c/hhp

[2] https://scikit-learn.org/stable/index.html



(a) Logistic Regression - AdultCensus (b) Random Forest - AdultCensus (c) MLP-AdultCensus (d) AdaBoost-AdultCensus

Figure 2: Trade-off between F1 and $\Delta_{DP}(\mathcal{A}, \mathbf{C}^U)$ on the AdultCensus dataset.

models. The fact that the trade-off behaviors of AL-based methods varying across downstream predicting backbones also verifies such a statement.

Moreover, we find in Figure 2 that the variant of FCRL-Obj3, which is the FCRL with the same fairness constraints as CUFRL, fails to achieve the best frontier parity but outperforms the other baselines significantly. This is probably because it uses $I(\mathbf{y};\mathbf{z}|\mathbf{c})$ as the objective of decoder for representation learning. Since in CUFRL, $\mathbf{c}$ contains much more attributes and can be more likely to be explicitly (included) or implicitly (correlated) related to $\mathbf{x}$, and therefore $\mathbf{z}$. Using such an objective reduces information for an expressive-enough representation unless making a compromise on the fairness objective. Despite that, the proposed objective (*i.e.,* Equation 3) is still effective enough to boost the performance of the FCRL framework.

In addition, we also observe that: 1) MIFR-AL can achieve relatively low disparity but at a cost of low accuracy. This is because its fairness constraint is constructed considering the input $\mathbf{x}$ and the representation, which will directly hurt the information retrieval and thus the accuracy if fairness is strengthened. 2) Adversarial-based approaches LAFTR-AL and AdvFgt-AL both give poor performance. We credit such a phenomenon that the quality of the adversarial classifier is highly dependent on the audit-learning algorithm, which is not transferable to downstream tasks.

4.3 Ablation Study

To verify the effectiveness of each component, we conduct ablation studies with variants and test the F1 and the corresponding $\Delta_{DP}(\mathcal{A}, \mathbf{C}^U)$ with β fixed as 0.05. The results on AdultCensus dataset with all four decision algorithm are demonstrated in Table 3, where Δ_{DP} denotes $\Delta_{DP}(\mathcal{A}, \mathbf{C}^U)$.

Without fairness-related objectives, **CUFRL w/o Izc** only serves as a simple classification-task pre-training encoder for the decision algorithm. Not surprisingly, it achieves high F1 but low $\Delta_{DP}(\mathcal{A}, \mathbf{C}^U)$. **CUFRL w/o Izx** denotes the variant of our approach that the information bottleneck term is removed. It can be found that the classification accuracy suffers a slight drop so as the fairness. Failing to obtain minimal sufficient information from data, during the process of debiasing, the variant is easier to be influenced by the fairness objective, which can be a conflict with the classification goal. As for **CUFRL w/o Izxc**, it directly predicts the task label but with the information bottleneck principle equipped. With no fairness regulations and better feature extraction ability,

Table 3: Ablation study.

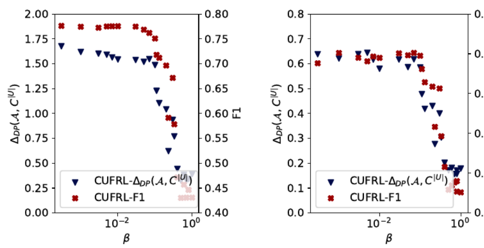
Method	MLP		Logistic Regression	
	F1	DP	F1	DP
CUFRL	0.7821	1.7459	0.7815	1.7273
CUFRL w/o Iyz	0.7790	1.7549	0.7808	1.7480
CUFRL w/o Izc	0.7807	1.8178	0.7817	1.6654
CUFRL w/o Izx	0.7559	1.7633	0.7610	1.7278
CUFRL w/o Izxc	0.7857	1.8925	0.7829	1.8117
Method	**AdaBoost**		**Random Forest**	
	F1	DP	F1	DP
CUFRL	0.7774	1.6608	0.7704	1.7041
CUFRL w/o Iyz	0.7701	1.7052	0.7727	1.7400
CUFRL w/o Izc	0.7716	1.8722	0.7756	1.7966
CUFRL w/o Izx	0.7798	1.7804	0.7758	1.7436
CUFRL w/o Izxc	0.7815	1.8791	0.7783	1.7982

it gets the best F1 but worst parity. There can be other objective functions for the classification task. For example, [13] uses $I(\mathbf{y};\mathbf{z}|\mathbf{c})$ to only infer $\mathbf{y}$ from $\mathbf{z}$ but with sensitive information eliminated. This could be an effective practice for a singular protected group, but we found from **CUFRL w/o Iyz** that it fails to reach as high F1 and as low parity as the proposed one. This can be resulted from that 1) such an objective can be reached by minimizing the $I(\mathbf{z};\mathbf{c})$ term, while itself, 2) giving an extra constraint on the classification task further reduces search space of $\mathbf{z}$. We observe similar trends on the Health Heritage dataset. To avoid redundancy, we omit the results here.

4.4 Parameter Analysis

4.4.1 Effect of β. β serves as a trade-off parameter in Equation 3. To evaluate the effect of β, we vary it within the range of $[1e{-}4, 1]$. The F1 and $\Delta_{DP}(\mathcal{A}, \mathbf{C}^U)$ *v.s.* β plots of the representation learning model on two datasets are depicted in Figure 3. The x axis is plotted in log-scale. It can be found that with β increases, better $\Delta_{DP}(\mathcal{A}, \mathbf{C}^U)$ can be obtained. Such a transition can be sensitive to β around 0.1.

4.4.2 Effect of Model Depth and Embedding Dimension. We now evaluate the influence of depth of the $I(\mathbf{z};\mathbf{x}|\mathbf{c}_i)$ estimation network and embedding dimension of representation. With β fixed as 0.05, the results on AdultCensus dataset are shown in Table 4. **G-{1,2}-D-{1,2}** denotes the depth of generator and discriminator, and d denotes



(a) AdultCensus (b) Health Heritage

Figure 3: Effect of β.

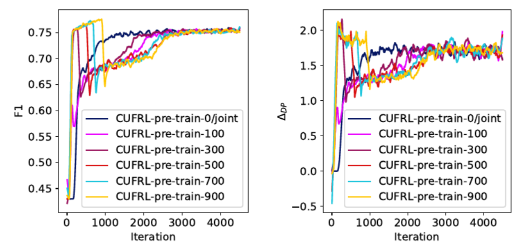
the embedding dimension of representations, which is chosen from {16, 32, 64}. We can observe that different decision algorithms have a different preference on the model parameters and **G**-{**1**}-**D**-{**1**} and d=16 is a relative fair choice that suits all. Similar results are observed on Health Heritage. To avoid redundancy, we omit the results here.

Table 4: Effect of model depth and embedding dimension.

Method		MLP		Logistic Regression	
Depth	d	F1	DP	F1	DP
G-1-C-1	16	0.7788	1.6657	0.7833	1.9117
	32	0.7733	1.6299	0.7854	1.8363
	64	0.7490	1.5115	0.7845	1.7719
G-2-C-1	16	0.7733	1.6861	0.7811	1.8587
	32	0.7669	1.5894	0.7826	1.7130
	64	0.7574	1.5910	0.7841	1.7426
G-1-C-2	16	0.7783	1.7510	0.7814	1.8665
	32	0.7597	1.5777	0.7808	1.8360
	64	0.7554	1.7128	0.7822	1.8390
G-2-C-2	16	0.7709	1.7062	0.7818	1.7661
	32	0.7637	1.5290	0.7823	1.8077
	64	0.7508	1.6850	0.7843	1.7857
Method		AdaBoost		Random Forest	
Depth	d	F1	DP	F1	DP
G-1-C-1	16	0.7805	1.9234	0.7772	1.8379
	32	0.7842	1.8874	0.7834	1.8160
	64	0.7807	1.7839	0.7811	1.7878
G-2-C-1	16	0.7767	1.8169	0.7706	1.8254
	32	0.7835	1.7599	0.7792	1.7685
	64	0.7786	1.7277	0.7806	1.7166
G-1-C-2	16	0.7816	1.8623	0.7796	1.8627
	32	0.7775	1.8492	0.7772	1.8198
	64	0.7781	1.864	0.7830	1.8793
G-2-C-2	16	0.7797	1.7807	0.7775	1.7739
	32	0.7811	1.7742	0.7799	1.7937
	64	0.7784	1.8255	0.7713	1.7848

4.5 Analysis on Training Paradigms

To further explore how the proposed approach works on different training paradigms and test if it is fine-tuning friendly, we conduct experiments on pre-trained features. The pre-trained representations are obtained with classification task only, *i.e.,* without the second term in Equation 3. To simulate representations of different expressiveness *w.r.t.* the downstream task, the number of [0, 100, 300, 500, 700, 900] iterations (batches) are implemented, where the expressiveness increases as the representation updates more times.



(a) F1 (b) $\Delta_{DP}(\mathcal{A}, \mathbf{C}^U)$

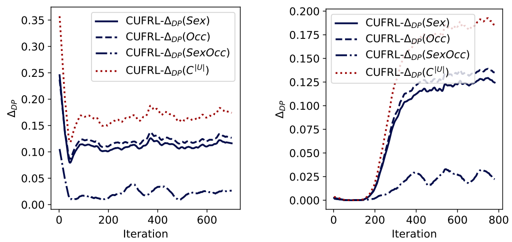
Figure 4: On AdultCensus dataset with different number of pre-training epochs.

The parity and F1 *v.s.* epoch plots on the training set of AdultCensus are reported in Figure 4. We can learn from plots that despite CUFRL needing more iterations to debias on more expressive representations, all runs converge to similar F1 and parity. This indicates that CUFRL not only works well in the case of joint-training but can be capable of debiasing on pre-trained feature representations, which implies a wider application scenario. Similar results are observed on Health Heritage. To avoid redundancy, we omit the results here.

4.6 Case Study: Look into individual sensitive attribute

We are interested in how the objective of minimizing $\Delta_{DP}(\mathcal{A}, \mathbf{C}^U)$ is related to minimizing the disparity on each sensitive attribute. To empirically investigate into this problem, we evaluate the representation learning model on a toy setting that only attributes "Sex" and "Occupation_Sales" (Occ) of the AdultCensus dataset are considered as singular sensitive attributes. That is, together with the one and only one level-1 sensitive attribute "Sex-Occupation_Sales" (SexOcc), we have $|\mathbf{C}^U| = 3$. We visualize the disparity *v.s.* iteration for each attribute (by calculation attribute-wise Δ_{DP}) and $\mathbf{C}^U$ in Figure 5. Results on two training paradigms are reported. To make the results comparable, we tune hyper-parameters to ensure the methods reach similar F1 $\sim$ 0.77 by the time of convergence.

We omit the pre-training epochs when plotting Figure 5(a). As depicted, initially, the disparity of all sensitive attributes possesses relatively large values (is true in our case but not necessary in every case, since a model doesn't necessarily bias toward every sensitive attribute) and tends to go down as training iteration increases. Though sometimes disparities *w.r.t.* individual sensitive attributes decrease/increase in a contrary/delayed trend to one another, they decline with the decrease of $\Delta_{DP}(\mathcal{A}, \mathbf{C}^U)$ in a generally smooth fashion. This verifies the effectiveness of the proposed $\Delta_{DP}(\mathcal{A}, \mathbf{C}^U)$ and objective function. As for joint-training based method, we can observe from Figure 5(b) that the overall change of disparity value is opposite to that of pre-training. Disparity starts at relatively low values and goes up as training iterations increase. This is because the model initializes as a random guesser and tends to have no bias at the beginning. Similar coherent optimization behavior between $\Delta_{DP}(\mathcal{A}, \mathbf{C}^U)$ and disparity of individual sensitive attributes can be observed. The synced convergence pattern verifies the effectiveness of minimizing $\Delta_{DP}(\mathcal{A}, \mathbf{C}^U)$ even when the sensitive attributes of interest are a subset of $\mathbf{C}^U$.



(a) Pre-training based (b) Joint-training based

Figure 5: Disparity *w.r.t.* toy individual attributes on Adult-Census dataset.

5 RELATED WORKS

5.1 Fairness in Machine Learning

According to the fairness notions, current machine learning fairness solutions mainly fall into three categories. 1) Group fairness [5, 13, 18, 40, 41, 46, 51] aims at treating different groups equally, where groups are defined as partitions of the population under a sensitive attribute. The widely used fairness definitions demographic parity [10] and equalized odds [15] belongs to this category. 2) Individual fairness [10, 29] targets providing similar results to similar individuals. 3) Subgroup fairness [16, 24, 26] intends to obtain similar outcomes between any subgroup and the whole population. Fair definitions in this category share similarities with group fairness but are modified to fit the subgroup *v.s.* entirety setting.

Various methods have been proposed for debiasing, based on the process phase, there are: 1) prep-process based [6, 7, 11, 20, 21, 55], which transforms or re-weights the training data to remove underlying discrimination; 2) in-processing based[2, 9, 23, 31, 45, 53, 54, 56], which adopts model tuning or regularization methods to tailor model training; and 3) post-processing based [8, 10, 22, 42], which processes the model output with certain algorithms or accesses external datasets that are not used for training for fairness. While these methods are confined to debiasing *w.r.t.* pre-defined protected groups, they shed light on designing models that are fair to all sensitive attributes and their combinations.

Our work falls into the categories of group fairness and preprocessing based.

5.2 Fair User Representation Learning

Representation learning, aiming at capturing generic prior of the data, has greatly boosted the success of machine learning algorithms. It has been widely applied in natural language processing, biological protein-protein network embedding, recommender systems, *etc.*. As a consequence of its ubiquity, ensuring the learned representations are fair is an essential task.

The majority of fair user representation learning approaches are studied under group fairness notions, where information-theoretic objectives [13, 48] and adversarial optimization strategies [17, 32, 36] are adapted to learn fair and invariant representations. More specifically, [48] proposes a controllable fair representation framework by using a dual optimization approach that considers both the model as well as Lagrange multipliers. [13] emphasizes the bounding fact of disparity and mutual information and provides a tight bound to regularize bias. [32] explores the connection between multiple group fairness notions to adversarial objectives. [17] proposes a unified framework for federated adversarial learning under the privacy consideration that users may not want to disclose their sensitive group variables. This shares a similar setting to our problem but is of intrinsically different concern due to the computational and privacy budget.

6 CONCLUSION AND FUTURE WORK

In this paper, we introduce a practical fair user representation learning setting, where the learned representation is expected to achieve parity over demographic groups defined by the universal set of sensitive attributes. To obtain fair and expressive representations, we propose the CUFRL method. In CUFRL, we derive an information-theoretical objective, where the disparity is shown to be bounded by mutual information terms. We also prove efficient reduction can be applied to reduce the computational complexity of fairness bounds. Thorough evaluations on two public real-world datasets demonstrate that compared with baselines, our method can robustly achieve good parity with reasonable classification accuracy on a wide range of downstream tasks (specifically, machine learning classification models).

The problem setting of universal fair representation learning introduced in this paper provides a natural guide to future research. For example, besides the notion of demographic parity used in this paper, there are many group-level fairness definitions also worth investigating *e.g.*, equalized odds and equal opportunity. Moreover, we consider universal fairness in a static setting, where all sensitive attributes need to be maintained at the same time. However, data providers and data users may cooperate in a query-based fashion, where debiased representations should be delivered according to unpredictable debiasing queries. In this case, the sensitive attribute set remains the universal set. But it remains a question of how to make query answering efficient and in-time, where adaptation/transfer techniques might be needed.

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A PROOFS

THEOREM 8. ***(Theorem 4 restated)*** *For $z, c_i \sim p(\mathbf{z}, \mathbf{c}_i)$, $z \in \mathbb{R}^d$, $\mathbf{c}_i \in \mathbf{C}^U$, $c_i \in \{1, ..., n_i\}$, and any decision algorithm $\mathcal{A}$ that acts on z, we have*

$$\sum_{i=1}^{|\mathbf{C}^U|} I(\mathbf{z}; \mathbf{c}_i) \geq g(\gamma, \Delta_{DP}(\mathcal{A}, \mathbf{C}^U))$$

where $\gamma = \min_i \alpha_i$, $\alpha_i = \min_j \pi_{i,j}$, $\pi_{i,j} = P(\mathbf{c}_i = j)$.

We prove Theorem 4 by first introducing the following Lemma:

LEMMA 9. *For some $z, c \sim p(\mathbf{z}, \mathbf{c})$, where c is any sensitive attribute $z \in \mathbb{R}^d$, $c \in \{1, ..., n\}$, and any decision algorithm $\mathcal{A}$ that acts on z, we have*

$$I(\mathbf{z}; \mathbf{c}) \geq \alpha^2 \Delta_{DP}(\mathcal{A}, \mathbf{c}))^2 \tag{10}$$

where $\alpha = \min \pi_i$, $\pi_i = P(\mathbf{c} = i)$.

Lemma 9 is heavily inspired by [13], which indicates that the mutual information bounds parity of any singular attribute.

The proof of Lemma 9 is offered as follows. Note that Theorem 4 can not be straightforwardly obtained by respectively adding the LHS and RHS of Equation 10 over i.

PROOF OF LEMMA 9. From the definition of $\Delta_{DP}(\mathcal{A}, \mathbf{c})$, we can get

$$\begin{aligned}
\Delta_{DP}(\mathcal{A}, \mathbf{c}) &= \max_{i,j} \; P(\hat{\mathbf{y}} = 1|\mathbf{c} = i) - P(\hat{\mathbf{y}} = 1|\mathbf{c} = j) \\
&= \max_{i,j} \int_z dz P(\hat{\mathbf{y}} = 1|\mathbf{z}) p(\mathbf{z}|\mathbf{c} = i) - P(\hat{\mathbf{y}} = 1|\mathbf{z}) p(\mathbf{z}|\mathbf{c} = j) \\
&\leq \max_{i,j} \int_z dz P(\hat{\mathbf{y}} = 1|\mathbf{z}) \; p(\mathbf{z}|\mathbf{c} = i) - p(\mathbf{z}|\mathbf{c} = j) \\
&\leq \max_{i,j} \int_z dz \; p(\mathbf{z}|\mathbf{c} = i) - p(\mathbf{z}|\mathbf{c} = j) \\
&= \max_{i,j} \; p(\mathbf{z}|\mathbf{c} = i) - p(\mathbf{z}|\mathbf{c} = j) \; = \max_{i,j} V\big(p(\mathbf{z}|\mathbf{c} = i), p(\mathbf{z}|\mathbf{c} = j)\big)
\end{aligned} \tag{11}$$

where $V(p_1, p_2) = \|p_1 - p_2\|$ is the variational distance between distribution p_1 and p_2. We can also calculate the mutual information between $\mathbf{z}$ and $\mathbf{c}$ as

$$\begin{aligned}
I(\mathbf{z}; \mathbf{c}) &= \mathbb{E}_{\mathbf{z},\mathbf{c}} log \frac{p(\mathbf{z}, \mathbf{c})}{p(\mathbf{z})p(\mathbf{c})} \\
&= \mathbb{E}_{\mathbf{z},\mathbf{c}} log \frac{p(\mathbf{z}|\mathbf{c})}{p(\mathbf{z})} \\
&= \sum_{i=1}^{N} \pi_i \mathbb{E}_{\mathbf{z}|\mathbf{c}=i} log \frac{p(\mathbf{z}|\mathbf{c} = i)}{p(\mathbf{z})} \\
&= \sum_{i=1}^{N} \pi_i KL\big(p(\mathbf{z}|\mathbf{c} = i) \| p(\mathbf{z})\big)
\end{aligned} \tag{12}$$

According to Pinsker's inequality [3], we know that

$$KL(p_1 \| p_2) \geq 2\|p_1 - p_2\|^2 = f(V) \tag{13}$$

Thus, we can derive that

$$\begin{aligned}
I(\mathbf{z}; \mathbf{c}) &= \sum_{i=1}^{N} \pi_i KL\big(p(\mathbf{z}|\mathbf{c} = i) \| p(\mathbf{z})\big) \\
&\geq \sum_{i=1}^{N} \pi_i f\Big(V\big(p(\mathbf{z}|\mathbf{c} = i), p(\mathbf{z})\big)\Big) \\
&\geq f\Big(\sum_{i=1}^{N} \pi_i V\big(p(\mathbf{z}|\mathbf{c} = i), p(\mathbf{z})\big)\Big) \\
&\geq f\big(\max_{i,j} \pi_i \|p(\mathbf{z}|\mathbf{c} = i) - p(\mathbf{z})\| + \pi_j \|p(\mathbf{z}|\mathbf{c} = j) - p(\mathbf{z})\|\big) \\
&\geq f\big(\alpha \max_{i,j} \|p(\mathbf{z}|\mathbf{c} = i) - p(\mathbf{z}|\mathbf{c} = j)\|\big) \\
&\geq f\big(\alpha \max_{i,j} |P(\hat{\mathbf{y}} = 1|\mathbf{c} = i) - P(\hat{\mathbf{y}} = 1|\mathbf{c} = j)|\big) \\
&= f\big(\alpha \Delta_{DP}(\mathcal{A}, \mathbf{c})\big)
\end{aligned} \tag{14}$$

f is convex function and also strictly increasing, non-negative. This completes the proof. □

Equipped with Lemma 9, we can then prove Theorem 4 as follows,

PROOF OF THEOREM 4. We can obtain from Equation 14 and the definition of $\Delta_{DP}(\mathcal{A}, \mathbf{C}^U)$ that,

$$\begin{aligned}
\sum_{i=1}^{|\mathbf{C}^U|} I(\mathbf{z}; \mathbf{c}_i) &= \sum_{i=1}^{|\mathbf{C}^U|} \sum_{j=1}^{N_i} \pi_{i,j} KL\big(p(\mathbf{z}|\mathbf{c}_i = j) \| p(\mathbf{z})\big) \\
&\geq \sum_{i=1}^{|\mathbf{C}^U|} f\big(\alpha \Delta_{DP}(\mathcal{A}, \mathbf{c}_i)\big) \\
&= \sum_{i=1}^{|\mathbf{C}^U|} {\alpha_i}^2 \Delta_{DP}(\mathcal{A}, \mathbf{c}_i)^2 \\
&\geq \gamma \sum_{i=1}^{|\mathbf{C}^U|} \Delta_{DP}(\mathcal{A}, \mathbf{c}_i)^2 \\
&= g(\gamma, \Delta_{DP}(\mathcal{A}, \mathbf{C}^U))
\end{aligned} \tag{15}$$

where $\gamma = \min_i \alpha_i^2$. $g(x, y) = xy^2$, which is strictly increasing, non-negative, and convex. This completes the proof. □

B PSEUDO CODE

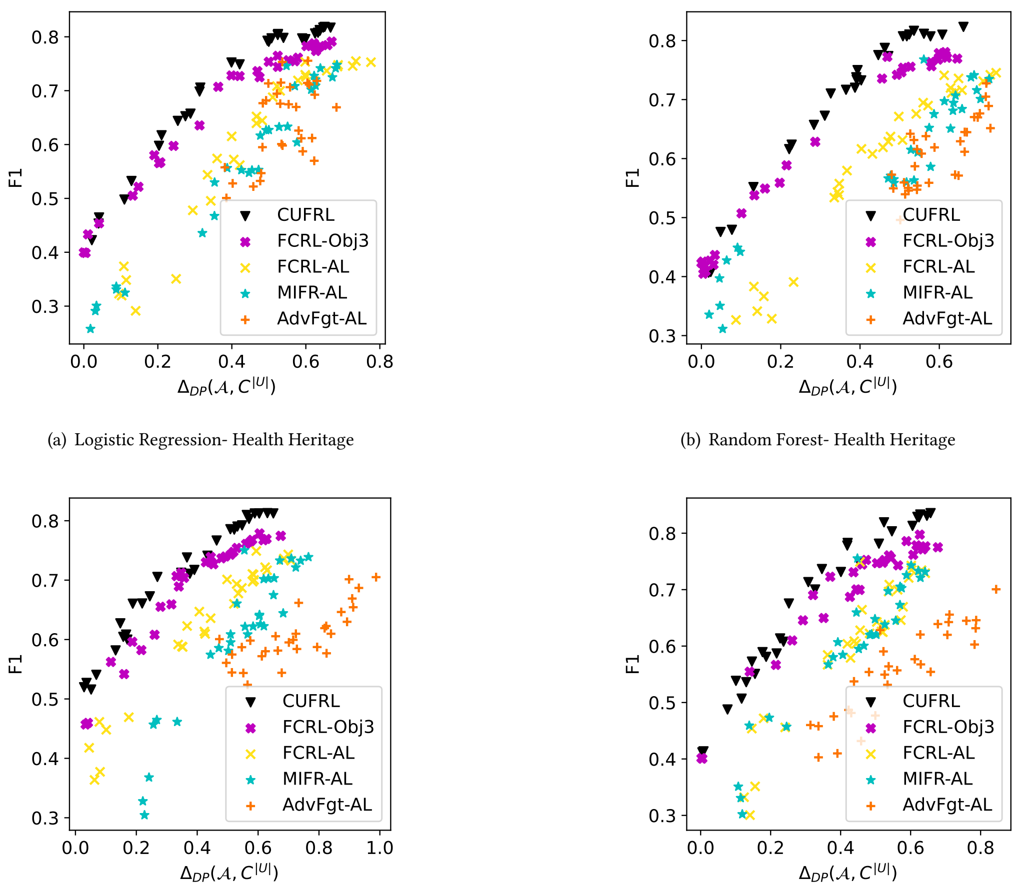
The pseudo code of computing L_c is presented in Algorithm 1.

C EXTRA RESULTS ON ACCURACY AND FAIRNESS TRADE-OFF ANALYSIS

Figure 6 shows the accuracy-fairness trade-off on Health-Heritage dataset. At similar parity, CUFRL achieves up-to ~3% higher F1 on the Health Heritage dataset than the best baseline.

Algorithm 1 Compute L_c

Input:
/* Assume representation **z** has been obtained in previous steps.*/
Batch data $\mathcal{B} = (\mathbf{z}, \mathbf{x}, \mathbf{c})$, inner iteration T;
Output: L_c;
for $t \in \{1, ..., T\}$ **do**
  Randomly split $\mathcal{B}$ equally to two parts $\mathcal{B}_{joint} = \{z_j, x_j, c_j\}_{j=0}^{n/2}$ and $\mathcal{B}_{gen} = \{z_j, x_j, c_j\}_{j=n/2}^{n}$;
  Compute $z'_j = NN_G(c_j) \in \mathbb{R}^{|C_1|}, \forall c_j \in \mathcal{B}_{joint}$ and obtain $\mathcal{B}_{prod} = \{z'_j, x_j, c_j\}_{j=0}^{n/2}$;
  Compute $L_{c_i,t}$ with $\{\mathcal{B}_{prod}, \mathbf{1}\}$, $\{\mathcal{B}_{prod}, \mathbf{0}\}$
Monte Carlo average $L_c = \sum_i Avg_t(L_{c_i,t})$



(a) Logistic Regression- Health Heritage

(b) Random Forest- Health Heritage

(c) MLP-Health Heritage

(d) AdaBoost-Health Heri.

Figure 6: Trade-off between F1 and $\Delta_{DP}(\mathcal{A}, C^U)$ on the Health Heritage dataset.